

How does HARM implement defence-in-depth?

The **HARM (Hallucination Assessment & Reduction Module)** implements a rigorous **defence-in-depth** strategy by forcing all unverified outputs from the Vision-Language Model (VLM) to pass through a series of **three independent, concentric validation layers**. Each layer acts as an isolated mathematical firewall targeting a specific type of AI systematic failure:

1. **Stage 1 (Schema Validation):** This layer acts as the outermost perimeter. It utilizes structural Pydantic parsing to ensure the raw VLM text response matches the required data types, format constraints, and enum variables. It catches format hallucinations and missing coordinate data.
2. **Stage 2 (Workspace Validation):** If the data structure passes Stage 1, it enters the physical perimeter. This layer checks all 3D waypoints against a geometric reachability sphere calculated directly from the robot's URDF model. It catches coordinate hallucinations by mapping out-of-bounds trajectory requests.
3. **Stage 3 (Kinematic Validation):** The final layer evaluates physical feasibility. It translates the spatial waypoints into joint paths to ensure that the required joint velocities and torques remain $\leq 80\%$ of the maximum hardware manufacturer boundaries. It blocks kinematic hallucinations and sudden velocity spikes before instructions reach the physical motors.

By decoupling these validations into independent steps, a failure or bypass in any single layer is automatically caught by the next downstream filter, satisfying the systematic capability requirements of **IEC 61508-3**.

What is the fail-safe state?

In compliance with **IEC 61508-1**, the **fail-safe state** is the predictable, conservative configuration the architecture defaults to whenever an anomalous input cannot be automatically resolved.

Within the HARM framework, the fail-safe state operates at two distinct levels:

- **Software Data Level:** If a VLM response violates the validation filters, HARM triggers a one-time error-suffix re-prompt loop. If it fails a second time, HARM injects a highly conservative, pre-verified fallback path command to protect software continuity.
- **Physical Hardware Level:** If a network drop, API crash, or unmitigated kinematic exception occurs, the system drops all active velocity profiles down to an immediate **HOLD position state**. The local control loop stops joint movement and safely locks the robot arm at its last known stable position, avoiding random or unguided kinetic motion.